

Take Home Activity 05

Student Number – 35040

Q1

```
1 public class HotelRoom {
2     int roomNumber;
3     String roomType;
4     double pricePerNight;
5     boolean isBooked;
6
7     HotelRoom() {
8         roomNumber = 0;
9         roomType = "Standard";
10        pricePerNight = 5000.0;
11        isBooked = false;
12    }
13
14    HotelRoom(int roomNumber, String roomType, double pricePerNight, boolean isBooked) {
15        this.roomNumber = roomNumber;
16        this.roomType = roomType;
17        this.pricePerNight = pricePerNight;
18        this.isBooked = isBooked;
19    }
20
21    double bookRoom(int nights) {
22        if (!isBooked) {
23            isBooked = true;
24            return pricePerNight * nights;
25        } else {
26            System.out.println("Room is already booked.");
27            return 0.0;
28        }
29    }
30
31    void checkoutRoom() {
32        isBooked = false;
33        System.out.println("Room " + roomNumber + " is now available.");
34    }
35
36    String getRoomSummary() {
37        return "Room " + roomNumber + " - " + roomType + " - Rs." + pricePerNight + " per night - Booked:" + isBooked;
38    }
39
40    public static void main(String[] args) {
41        HotelRoom room1 = new HotelRoom();
42        HotelRoom room2 = new HotelRoom(101, "Deluxe", 8500.0, false);
43
44        double cost = room1.bookRoom(3);
45        System.out.println("Total cost: " + cost);
46
47        room1.bookRoom(2);
48
49        room1.checkoutRoom();
50
51        double newCost = room1.bookRoom(2);
52        System.out.println("New total cost: " + newCost);
53
54        System.out.println(room1.getRoomSummary());
55        System.out.println(room2.getRoomSummary());
56    }
57 }
58
59
60
61
62
63
64
65
66
```

```

public class HotelRoom {
    int roomNumber;
    String roomType;
    double pricePerNight;
    boolean isBooked;

    HotelRoom() {
        roomNumber = 0;
        roomType = "standard";
        pricePerNight = 5000.0;
        isBooked = false;
    }

    HotelRoom(int roomNumber, String roomType, double pricePerNight, boolean isBooked) {
        this.roomNumber = roomNumber;
        this.roomType = roomType;
        this.pricePerNight = pricePerNight;
        this.isBooked = isBooked;
    }

    double bookRoom(int nights) {
        if (!isBooked) {
            isBooked = true;
            return pricePerNight * nights;
        } else {
            System.out.println("Room is already booked.");
            return 0.0;
        }
    }
}

```

```

    void checkoutRoom() {
        isBooked = false;
        System.out.println("Room " + roomNumber + " is now available.");
    }

    String getRoomSummary() {
        return "Room " + roomNumber + " " + roomType + " - Rs. " +
            pricePerNight + " per Night - Booked: " + isBooked;
    }
}

```

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```

public static void main(String[] args) {
    HotelRoom room1 = new HotelRoom();
    HotelRoom room2 = new HotelRoom(101, "Deluxe", 8500.0);

    double cost = room1.bookRoom(3);
    System.out.println("Total cost: " + cost);

    room1.bookRoom(2);
    room1.checkoutRoom();

    double newcost = room1.bookRoom(2);
    System.out.println("New total cost: " + newcost);

    System.out.println(room1.getRoomSummary());
    System.out.println(room2.getRoomSummary());
}

```

```
Total cost: 15000.0
Room is already booked.
Room0 is now available.
New total cost: 10000.0
Room0-Standard- Rs.5000.0per night - Booked:true
Room101-Deluxe- Rs.8500.0per night - Booked:false
```

Q2

```
1 public class OnlineOrder {
2     int orderId;
3     String customerName;
4     double itemPrice;
5     int quantity;
6
7     public OnlineOrder(int orderId, String customerName, double itemPrice, int quantity) {
8         this.orderId = orderId;
9         this.customerName = customerName;
10        this.itemPrice = itemPrice;
11        this.quantity = quantity;
12    }
13
14    public double calculateTotal(double discountPercentage) {
15        double total = itemPrice * quantity;
16        double discount = total * discountPercentage / 100;
17        return total - discount;
18    }
19
20    public void updateQuantity(int newQuantity) {
21        if (newQuantity > 0) {
22            quantity = newQuantity;
23            System.out.println("Quantity updated to " + newQuantity + ".");
24        } else {
25            System.out.println("Invalid quantity.");
26        }
27    }
28
29    public String getOrderSummary() {
30        return "Order ID: " + orderId +
31            ", Customer Name: " + customerName +
32            ", Item Price: Rs. " + itemPrice +
33            ", Quantity: " + quantity;
34    }
35
36    public static void main(String[] args) {
37        OnlineOrder order1 = new OnlineOrder(1, "Nimal", 2500.0, 2);
38        OnlineOrder order2 = new OnlineOrder(2, "Kamal", 1500.0, 3);
39
40        double finalAmount = order1.calculateTotal(10);
41        System.out.println("Final amount after 10% discount: Rs. " + finalAmount);
42
43        order1.updateQuantity(5);
44        order1.updateQuantity(-2);
45
46        System.out.println(order1.getOrderSummary());
47        System.out.println(order2.getOrderSummary());
48    }
49 }
```

```

public class OnlineOrder {
    int orderId;
    String customerName;
    double itemPrice;
    int quantity;

    public OnlineOrder(int orderId, String customerName, ...) {
        this.orderId = orderId;
        this.customerName = customerName;
        this.itemPrice = itemPrice;
        this.quantity = quantity;
    }
}

```

Allex

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```

public double Calculation(double discount) {
    double total = itemPrice * quantity;
    double discount = total * discount / 100;
    return total - discount;
}

```

}

```

public void updateQuantity(int newQuantity) {
    if (newQuantity > 0) {
        quantity = newQuantity;
        System.out.println("Quantity updated to: " + quantity);
    } else {
        System.out.println("Invalid quantity");
    }
}

```

```

public String getOrderSummary() {
    return "Order ID: " + orderId + ", Customer Name: " +
        customerName + ", Item Price: Rs. " + itemPrice + " and " +
        quantity;
}

```

```

public static void main(String[] args) {
    OnlineOrder ord1 = new OnlineOrder(1, "Alice", 100, 2);
    OnlineOrder ord2 = new OnlineOrder(2, "Bob", 150, 3);

    double finalAmount = ord1.calculateTotal(10);
    System.out.println("Final Amount: Rs. " + finalAmount);

    ord1.updateQuantity(5);
    ord2.updateQuantity(-2);
}

```

```

System.out.println("Order Summary: " + ord1.getOrderSummary());
System.out.println("Order Summary: " + ord2.getOrderSummary());
}

```

Allex

Final amount after 10% discount: Rs. 4500.0

Quantity updated to 5.

Invalid quantity.

Order ID: 1, Customer Name: Nimal, Item Price: Rs. 2500.0, Quantity: 5

Order ID: 2, Customer Name: Kamal, Item Price: Rs. 1500.0, Quantity: 3

Q3

```
1 public class StudentResult {
2     int studentId;
3     String studentName;
4     double mathMark;
5     double scienceMark;
6     double englishMark;
7
8     public StudentResult(int studentId, String studentName, double mathMark, double scienceMark, double englishMark) {
9         this.studentId = studentId;
10        this.studentName = studentName;
11        this.mathMark = mathMark;
12        this.scienceMark = scienceMark;
13        this.englishMark = englishMark;
14    }
15
16    public double calculateAverage() {
17        return (mathMark + scienceMark + englishMark) / 3;
18    }
19
20    public String getGrade() {
21        double average = calculateAverage();
22
23        if (average >= 75) {
24            return "A";
25        } else if (average >= 65) {
26            return "B";
27        } else if (average >= 50) {
28            return "C";
29        } else if (average >= 35) {
30            return "S";
31        } else {
32            return "F";
33        }
34    }
35
36    public void updateMark(String subject, double newMark) {
37        if (subject.equalsIgnoreCase("math")) {
38            mathMark = newMark;
39        } else if (subject.equalsIgnoreCase("science")) {
40            scienceMark = newMark;
41        } else if (subject.equalsIgnoreCase("english")) {
42            englishMark = newMark;
43        } else {
44            System.out.println("Invalid subject name.");
45        }
46    }
47
48    public String getResultSheet() {
49        return "Student ID: " + studentId +
50            "\nStudent Name: " + studentName +
51            "\nMath Mark: " + mathMark +
52            "\nScience Mark: " + scienceMark +
53            "\nEnglish Mark: " + englishMark +
54            "\nAverage: " + calculateAverage() +
55            "\nGrade: " + getGrade();
56    }
57
58    public static void main(String[] args) {
59        StudentResult student1 = new StudentResult(1, "Amal", 80, 75, 70);
60        StudentResult student2 = new StudentResult(2, "Saman", 60, 55, 50);
61
62        student1.updateMark("math", 90);
63        student1.updateMark("history", 70);
64
65        System.out.println(student1.getResultSheet());
66        System.out.println();
67
68        System.out.println(student2.getResultSheet());
69    }
70 }
```



```

public class StudentResult {
    int studentID;
    String studentName;
    double mathMark;
    double scienceMark;
    double englishMark;

    public StudentResult(int studentID, String studentName,
        double mathMark, double scienceMark, double englishMark) {
        this.studentID = studentID;
        this.studentName = studentName;
        this.mathMark = mathMark;
        this.scienceMark = scienceMark;
        this.englishMark = englishMark;
    }

    public double calculateAverage() {
        return (mathMark + scienceMark + englishMark) / 3;
    }

    public String getGrade() {
        double average = calculateAverage();

        if (average >= 75) {
            return "A";
        } else if (average >= 65) {
            return "B";
        } else if (average >= 50) {
            return "C";
        } else if (average >= 35) {
            return "D";
        } else {
            return "F";
        }
    }
}

```

```

public void updateMark(String subject, double mark) {
    if (subject.equalsIgnoreCase("math")) {
        mathMark = mark;
    } else if (subject.equalsIgnoreCase("science")) {
        scienceMark = mark;
    } else {
        System.out.println("Invalid Subject Name");
    }
}

public String getResultSheet() {
    return "Student ID: " + studentID + " Student Name: " + studentName +
        " Math Mark: " + mathMark + " Science Mark: " + scienceMark +
        " English Mark: " + englishMark + " Average: " + calculateAverage() +
        " Grade: " + getGrade();
}

public static void main(String[] args) {
    StudentResult student1 = new StudentResult(1, "Alice", 85, 78, 92);
    StudentResult student2 = new StudentResult(2, "Bob", 72, 65, 80);

    student1.updateMark("math", 90);
    student1.updateMark("history", 70);

    System.out.println(student1.getResultSheet());
    System.out.println(student2.getResultSheet());
}

```

Math Mark: 90.0
Science Mark: 75.0
English Mark: 70.0
Average: 78.33333333333333
Grade: A

Student ID: 2
Student Name: Saman
Math Mark: 60.0
Science Mark: 55.0
English Mark: 50.0
Average: 55.0
Grade: C

Q4

```
1 public class VehicleRental {
2     int vehicleId;
3     String vehicleType;
4     double ratePerDay;
5     double availableKm;
6     boolean isAvailable;
7
8     public VehicleRental() {
9         vehicleId = 0;
10        vehicleType = "Unknown";
11        ratePerDay = 0.0;
12        availableKm = 0.0;
13        isAvailable = true;
14    }
15
16    public VehicleRental(int vehicleId, String vehicleType, double ratePerDay, double availableKm, boolean isAvailable) {
17        this.vehicleId = vehicleId;
18        this.vehicleType = vehicleType;
19        this.ratePerDay = ratePerDay;
20        this.availableKm = availableKm;
21        this.isAvailable = isAvailable;
22    }
23
24    public String rentVehicle(int days, double estimatedKm) {
25        if (!isAvailable) {
26            return "Vehicle is not available.";
27        } else if (estimatedKm > availableKm) {
28            return "Estimated kilometers exceed available kilometers.";
29        } else {
30            isAvailable = false;
31            double totalCost = ratePerDay * days;
32            return "Vehicle rented successfully. Total cost: Rs. " + totalCost;
33        }
34    }
35
36    public void returnVehicle(double kmUsed) {
37        availableKm = availableKm - kmUsed;
38        isAvailable = true;
39        System.out.println("Vehicle returned. Available Km: " + availableKm + ", Available: " + isAvailable);
40    }
41
42    public String getVehicleInfo() {
43        return "Vehicle ID: " + vehicleId +
44            ", Vehicle Type: " + vehicleType +
45            ", Rate Per Day: Rs. " + ratePerDay +
46            ", Available Km: " + availableKm +
47            ", Is Available: " + isAvailable;
48    }
49
50    public static void main(String[] args) {
51        VehicleRental vehicle1 = new VehicleRental();
52        VehicleRental vehicle2 = new VehicleRental(101, "Car", 6000.0, 500.0, true);
53
54        System.out.println(vehicle2.getVehicleInfo());
55
56        System.out.println(vehicle2.rentVehicle(3, 200));
57        System.out.println(vehicle2.getVehicleInfo());
58
59        System.out.println(vehicle2.rentVehicle(2, 100));
60
61        vehicle2.returnVehicle(150);
62        System.out.println(vehicle2.getVehicleInfo());
63
64        System.out.println(vehicle2.rentVehicle(2, 100));
65        System.out.println(vehicle2.getVehicleInfo());
66    }
67 }
```

```

public class VehicleRental {
    int vehicleID;
    String vehicleType;
    double ratePerDay;
    double availableKm;
    boolean isAvailable;

    public VehicleRental() {
        vehicleID = 0;
        vehicleType = "Unknown";
        ratePerDay = 0.0;
        available = 0.0;
        isAvailable = true;
    }

    public VehicleRental(int vehicleID, String vehicleType,
        double ratePerDay, double availableKm, boolean isAvailable) {
        this.vehicleID = vehicleID;
        this.vehicleType = vehicleType;
        this.ratePerDay = ratePerDay;
        this.availableKm = availableKm;
        this.isAvailable = isAvailable;
    }

    public String rentVehicle(int days, double estimateKm) {
        if (!isAvailable) {
            return "Vehicle is not available.";
        } else if (estimateKm > availableKm) {
            return "Estimated kilometers exceed available km.";
        } else {
            isAvailable = false;
            double totalCost = ratePerDay * days;
            return "Vehicle rented successfully. Total cost: " + totalCost;
        }
    }

    public void returnVehicle(double kmUsed) {
        isAvailable = true;
        system.out.println("Vehicle returned, Available km: " + availableKm);
    }
}

```

```

    public void returnVehicle(double kmUsed) {
        availableKm = availableKm - kmUsed;
        isAvailable = true;
        system.out.println("Vehicle returned, Available km: " + availableKm);
    }
}

```

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```

    public String getVehicleInfo() {
        return "Vehicle ID: " + vehicleID + ", Vehicle Type: " + vehicleType + "Rate per Day: Rs " + ratePerDay + "Available km: " + availableKm + "is Available: " + isAvailable;
    }

    public static void main(String[] args) {
        VehicleRental vehicle1 = new VehicleRental(1, "Car", 100, 100, true);
        VehicleRental vehicle2 = new VehicleRental(2, "Bike", 50, 50, true);

        system.out.println(vehicle1.getVehicleInfo());
        system.out.println(vehicle2.getVehicleInfo());
        system.out.println(vehicle1.rentVehicle(3, 200));
        system.out.println(vehicle1.getVehicleInfo());
        system.out.println(vehicle2.rentVehicle(2, 100));
        system.out.println(vehicle2.getVehicleInfo());
        system.out.println(vehicle1.returnVehicle(150));
        system.out.println(vehicle1.getVehicleInfo());
    }
}

```



```

Vehicle ID: 101, Vehicle Type: Car, Rate Per Day: Rs. 6000.0, Available Km: 500.0, Is Available: true
Vehicle rented successfully. Total cost: Rs. 18000.0
Vehicle ID: 101, Vehicle Type: Car, Rate Per Day: Rs. 6000.0, Available Km: 500.0, Is Available: false
Vehicle is not available.
Vehicle returned. Available Km: 350.0, Available: true
Vehicle ID: 101, Vehicle Type: Car, Rate Per Day: Rs. 6000.0, Available Km: 350.0, Is Available: true
Vehicle rented successfully. Total cost: Rs. 12000.0
Vehicle ID: 101, Vehicle Type: Car, Rate Per Day: Rs. 6000.0, Available Km: 350.0, Is Available: false

```

Q5

```

1 public class LibraryBook {
2     int bookId;
3     String title;
4     String author;
5     int totalCopies;
6     int borrowedCopies;
7
8     public LibraryBook() {
9         bookId = 0;
10        title = "Unknown";
11        author = "Unknown";
12        totalCopies = 0;
13        borrowedCopies = 0;
14    }
15
16    public LibraryBook(int bookId, String title, String author, int totalCopies) {
17        this.bookId = bookId;
18        this.title = title;
19        this.author = author;
20        this.totalCopies = totalCopies;
21        this.borrowedCopies = 0;
22    }
23
24    public String borrowBook(int count) {
25        int available = totalCopies - borrowedCopies;
26
27        if (count <= available) {
28            borrowedCopies = borrowedCopies + count;
29            available = totalCopies - borrowedCopies;
30            return "Successfully borrowed " + count + " copy/copies. Remaining: " + available;
31        } else {
32            return "Only " + available + " copy/copies available.";
33        }
34    }
35
36    public void returnBook(int count) {
37        if (count <= borrowedCopies) {
38            borrowedCopies = borrowedCopies - count;
39            System.out.println(count + " copy/copies returned successfully.");
40        } else {
41            System.out.println("Return count exceeds borrowed copies.");
42        }
43    }
44
45    public int getAvailableCopies() {
46        return totalCopies - borrowedCopies;
47    }
48
49    public String getBookInfo(String shelf) {
50        return "Book ID: " + bookId +
51            ", Title: " + title +
52            ", Author: " + author +
53            ", Total Copies: " + totalCopies +
54            ", Borrowed Copies: " + borrowedCopies +
55            ", Shelf: " + shelf;
56    }
57
58    public static void main(String[] args) {
59        LibraryBook book2 = new LibraryBook(101, "Java Programming", "James Gosling", 5);
60
61        System.out.println(book2.getBookInfo("A1"));
62
63        System.out.println(book2.borrowBook(3));
64        System.out.println(book2.getBookInfo("A1"));
65
66        System.out.println(book2.borrowBook(4));
67
68        book2.returnBook(2);
69        System.out.println(book2.getBookInfo("A1"));
70
71        System.out.println(book2.borrowBook(2));
72        System.out.println(book2.getBookInfo("A1"));
73    }
74 }

```

```
public class LibraryBook {
```

```
    int bookId;  
    String title;  
    String author;  
    int totalCopies;  
    int borrowedCopies;
```

```
    public LibraryBook() {
```

```
        bookId = 0;  
        title = "unknown";  
        author = "unknown";  
        totalCopies = 0;  
        borrowedCopies = 0;
```

```
}
```

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```
    public LibraryBook(int bookId, String title, String author) {
```

```
        this.bookId = bookId;  
        this.title = title;  
        this.author = author;  
        this.totalCopies = totalCopies;  
        this.borrowedCopies = 0;
```

```
}
```

```
    public String borrowBook(int count) {
```

```
        int available = totalCopies - borrowedCopies;
```

```
        if (count <= available) {
```

```
            borrowedCopies = borrowedCopies + count;  
            available = totalCopies - borrowedCopies;  
            return "successfully borrowed" + count + "  
            copy/copies. Rem : " + available;
```

```
        } else {
```

```
            return "only" + available + "copy/copies
```

```
        }
```

```
    }  
    public void returnBook(int count) {
```

```
        if (count <= borrowedCopies) {
```

```
            borrowedCopies = borrowedCopies - count;  
            System.out.println(count + "copy/copies return
```

```
        } else {
```

```
            System.out.println("Return count bow ");
```

```
        }
```

```
    }  
    public int getAvailable() {
```

```
        return totalCopies - borrowedCopies;
```

```
Book ID: 101, Title: Java Programming, Author: James Gosling, Total Copies: 5, Borrowed Copies: 0, Shelf: A1
Successfully borrowed 3 copy/copies. Remaining: 2
Book ID: 101, Title: Java Programming, Author: James Gosling, Total Copies: 5, Borrowed Copies: 3, Shelf: A1
Only 2 copy/copies available.
2 copy/copies returned successfully.
Book ID: 101, Title: Java Programming, Author: James Gosling, Total Copies: 5, Borrowed Copies: 1, Shelf: A1
Successfully borrowed 2 copy/copies. Remaining: 2
Book ID: 101, Title: Java Programming, Author: James Gosling, Total Copies: 5, Borrowed Copies: 3, Shelf: A1
```